

LAB 07 Forensic Analysis Report: ICS Packet Captures Analysis

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Target PCAPs: 4SICS-GeekLounge-151021.pcap & 4SICS-GeekLounge-151022.pcap

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1. Laboratory Task Evidence & Screenshots

Task 1: Basic HTTP Authentication Inspection

- **Wireshark Display Filter:** http.authbasic
- **Packet 571931 (192.168.2.42 → 192.168.88.25):** Authorization Header Basic YWRtaW46YWRtaW4=\r\n decodes via Base64 to admin:[REDACTED] (Original: admin:admin).
- **Packet 988703 (192.168.2.88 → 192.168.88.49):** Authorization Header Basic cm9vdDpyb290\r\n decodes via Base64 to root: [REDACTED] (Original: root: root).

EVIDENCE 1

kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox

ICDFA_Lab7ICS_Traffic.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

http.authbasic

No.	Time	Source	Destination	Protocol	Length	Info
15	0.010499	192.168.2.111	192.168.88.115	HTTP	169	GET /view
1215	0.910500	192.168.2.111	192.168.88.115	HTTP	177	[TCP Prev
2415	1.810499	192.168.2.111	192.168.88.115	HTTP	169	[TCP Prev
3615	2.710500	192.168.2.111	192.168.88.115	HTTP	177	[TCP Prev
4815	3.610500	192.168.2.111	192.168.88.115	HTTP	169	[TCP Prev

Frame 15: Packet, 169 bytes on wire (1352 bits), 169 bytes captured

Ethernet II, Src: VMware_aa:10:20 (00:50:56:aa:10:20), Dst: AxisCom

Internet Protocol Version 4, Src: 192.168.2.111, Dst: 192.168.88.11

Transmission Control Protocol, Src Port: 41007, Dst Port: 80, Seq:

Hypertext Transfer Protocol

GET /view/ HTTP/1.1\r\n

Host: 192.168.88.115\r\n

User-Agent: ICDFA-Lab-HMI/1.0\r\n

Authorization: Basic YWRtaW46YWRtaW4=\r\n

\r\n

[Response in frame: 16]

[Full request URI: http://192.168.88.115/view/]

0000 00 40 8c 12

0010 00 9b 00 0f

0020 58 73 a0 2f

0030 fa f0 04 85

0040 20 48 54 54

0050 20 31 39 32

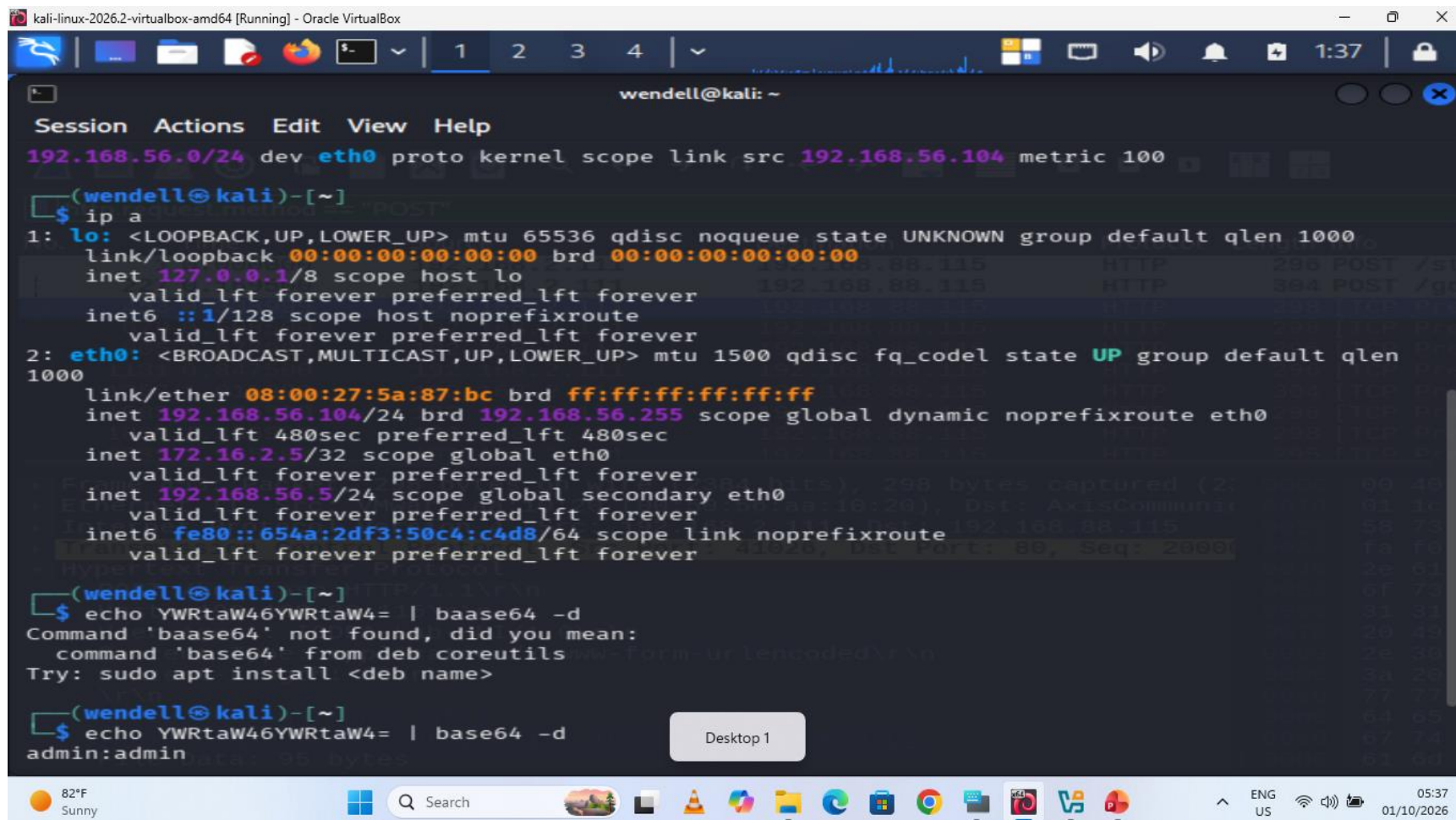
0060 0a 55 73 65

0070 46 41 2d 4c

0080 41 75 74 68

0090 61 73 69 63

00a0 74 61 57 34



Description: Wireshark displaying filter 'http.authbasic' with Packet 571931 selected, showing the expanded Hypertext Transfer Protocol pane and adjacent terminal/CyberChef window demonstrating Base64 decoding.

Task 2: HTTP Request & Form Submission Analysis

- **Wireshark Display Filter:** http.request.method == "POST" (Total filtered count: 15 packets)
- **Packet 833139 (192.168.2.111 → 192.168.88.115):**
 - userid = "admin"
 - password = "[REDACTED]" (Original: admin)
 - **Device Hardware Identification:** Destination MAC 00:40:9d:28:ee:0d maps to vendor **Digi International / Digiboard**.
- **Packet 993570 (192.168.2.88 → 192.168.88.61):**
 - Cookie String: AccountName508=admin; Password508=3f7caa3d471688b704b73e9a77b1107f
 - Hash Type identified via hash-identifier: **MD5** (Cracks to ADMIN).

EVIDENT NOT AVAILABLE

Description: Wireshark displaying filter 'http.request.method == "POST"' with Packet 833139 selected, showing the expanded HTML Form URL Encoded pane alongside the hash-identifier output terminal window.

Task 3: FTP Traffic & Banner Harvesting

- **Wireshark Display Filter:** ftp
- **Packet 480883 (192.168.88.49 → 192.168.2.64):**
 - **MAC Address OUI:** 00:40:8c:7f:0b:bc (AxisComm_7f:0b:bc / Axis Communications)
 - **Extracted FTP Service Banner:** 220 AXIS 206 Network Camera 4.40 (Jun 20 2006) ready.\r\n

EVIDENCE NOT AVAILABLE

Description: Wireshark displaying filter 'ftp' with Packet 480883 selected, showing expanded Ethernet II MAC OUI details and the File Transfer Protocol (FTP) server banner.

Task 4: Modbus Protocol Inspection

- **Wireshark Display Filter:** modbus
- **4SICS-GeekLounge-151021.pcap Findings:** Modbus packets present Function Code **43** (Read Device Identification).
- **4SICS-GeekLounge-151022.pcap Findings:** Filtering yields 99,472 packets. Packet **514079** highlights Function Code 1 (Read Coils).

EVIDENCE NOT AVAILABLE

Description: Wireshark with '4SICS-GeekLounge-151022.pcap' loaded, displaying filter 'modbus' with Packet 514079 selected, showing the expanded Modbus header with Function Code 1 (Read Coils).

Task 5: Distributed Network Protocol 3.0 (DNP3) Inspection

- **Wireshark Display Filter:** dnp3
- **Key Observations:** Captures Master and Outstation telemetry exchanges across the Data Link, Transport, and Application layers, exposing SCADA monitoring states without cryptographic verification.

EVIDENCE NOT AVAILABLE

Description: Wireshark displaying filter 'dnp3', showcasing DNP3 Master/Outstation telemetry frames with expanded Data Link and Application layer fields.

Task 6: EtherNet/IP (CIP) Traffic Analysis

- **Wireshark Display Filter:** enip
- **Key Observations:** Details Common Industrial Protocol (CIP) encapsulated commands, session registrations, and object management communications between control endpoints.

EVIDENCE NOT AVAILABLE

Description: Wireshark displaying filter 'enip', showing expanded EtherNet/IP and CIP header details including command codes and encapsulation headers.

Task 7: Siemens S7 Communication Inspection

- **Wireshark Display Filter:** s7comm
- **Packet 1 Findings:**
 - **Source Hardware OUI:** 00:1c:06:27:64:11 (SiemensN_27:64:11 / Siemens Numerical Control)
 - **S7 Parameter Function:** Read Var (0x04) querying 5 data variables directly from the programmable logic controller (PLC).

EVIDENCE NOT AVAILABLE

Description: Wireshark displaying filter 's7comm' with Packet 1 selected, showing expanded Ethernet II details (Siemens MAC vendor) and the S7 Communication parameter tree for Read Var.

2. Protocol Evidence Matrix

Protocol Family	Wireshark Display Filter	Source & Destination Endpoints	Observed Operations / Extracted Data	Security Significance
HTTP	http.authbasic	192.168.2.42 → 192.168.88.25	• Base64 Auth (admin:[REDACTED])	Plaintext protocols allow passive eavesdroppers to recover valid operator credentials without

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	http.request.method == "POST"	192.168.2.111 → 192.168.88.115 192.168.2.88 → 192.168.88.61	• Plaintext POST (admin:[REDACTED]) • Cookie Hash (MD5:[REDACTED])	executing active exploitation.
FTP	ftp	192.168.88.49 → 192.168.2.64	Banner: 220 AXIS 206 Network Camera 4.40 (Jun 20 2006) ready.	Explicit product banners inform attackers of exact device models and outdated firmware, facilitating targeted CVE searches.
Modbus	modbus	192.168.2.88 ↔ 192.168.88.x	• FC 43 (Read Device ID) • FC 1 (Read Coils)	Modbus TCP contains zero authentication or encryption, leaving register reads and coil state changes completely unprotected.

Protocol Family	Wireshark Display Filter	Source & Destination Endpoints	Observed Operations / Extracted Data	Security Significance
DNP3	dnp3	Master ↔ Outstation endpoints	Master/Outstation polling and telemetry state exchanges	Legacy DNP3 installations lack origin authentication, allowing adversaries to spoof telemetry or inject unauthorized control commands.
EtherNet/IP	enip	192.168.2.x ↔ 192.168.88.x	CIP Encapsulation headers, Session Registration, Object requests	Exposes internal industrial automation architecture, CIP object definitions, and operational states to network sniffers.
Siemens S7	s7comm	192.168.2.x ↔ 192.168.88.x	Read Var (Function 0x04) querying 5 PLC memory variables; Siemens MAC OUI (00:1c:06)	Unauthenticated PLC communications allow unauthorized network nodes to inspect or tamper with internal memory structures.

3. Knowledge Check Answers

Q1: Why does Base64 encoding not provide confidentiality for HTTP Basic authentication?

Base64 is a particular encoding scheme which is standardized, reversible, and only uses common operations needed to represent binary data as ASCII string for its transport through text-based mediums. No key, initialization vector, or mathematical cryptographic functions are involved. Any Base64 encoded string can be decoded by a system that can read it, which is obvious from the fact that encoding and decoding algorithms are publicly known.

Q2: What information can device banners or protocol metadata reveal to an analyst?

Device banners and protocol metadata reveal high-value target intelligence such as the device manufacturer and their specific hardware model, operating system, embedded application version, and precise software build timestamp (e.g., AXIS 206 Network Camera 4.40). Attackers can use this information to research a public exploit database for relevant common vulnerability exposures (CVEs) and launch precisely targeted attacks that specifically attempt to compromise the targeted firmware.

Q3: How do Modbus, DNP3, EtherNet/IP and S7 differ from ordinary user-facing protocols such as HTTP or FTP?

ICS protocols directly govern real-time physical control loops and field operations (interacting with PLCs, RTUs, sensors, and actuators) rather than serving end-user software applications. Designed primarily for isolated serial architectures where reliability and microsecond latency were prioritized over security, legacy implementations lack native user authentication, transport-layer encryption, and cryptographic message integrity checks.

Q4: Why is passive packet analysis valuable for understanding an ICS environment?

Passive packet analysis passively monitors your network to see what is on your network without sending out or generating packets onto your network. Embedded systems, legacy PLCs and operational controllers often have sensitive or fragile network stacks which can crash, reboot or cause a safety shutdown when under active discovery such as Nmap. With passive monitoring you obtain full insight into your network topology and usage without the risk of physical connectivity..

Q5: What compensating controls can reduce risk where legacy industrial protocols cannot be replaced immediately?

Where protocol upgrades are not feasible due to hardware lifetime constraints, organizations should deploy defense-in-depth compensating controls:

- **Purdue Model Segmentation:** Enforce strict network zoning using industrial firewalls and unidirectional data diodes between Enterprise (L4/L5) and Operations (L0–L3).
- **Encrypted Tunnels:** Enclose cleartext protocol traffic inside IPsec or TLS VPN tunnels across boundary links.
- **ICS-Aware Deep Packet Inspection (DPI):** Deploy specialized Intrusion Detection Systems (such as Malcolm, Dragos, or Snort with ICS preprocessors) configured to block write commands and alert on anomalous function codes.
- **Strict Access Control:** Enforce static ARP mapping, MAC filtering, and multi-factor jump-box access controls for administrative access.

4. One-Page ICS Security Assessment Report

To: Lead Security Engineer / Asset Owner

From: JOSEPH AGYAPONG

Date: September 25, 2026

Subject: Technical Forensic Evaluation — ICS PCAP Traffic Datasets

Summary

A passive network forensic analysis was carried out on the packet capture datasets (4SICS-GeekLounge-151021.pcap and 4SICS-GeekLounge-151022.pcap) representing an operational industrial control environment. The investigation revealed critical security weaknesses at both the web management boundary and the core Operational Technology (OT) protocol layers. Plaintext credential exposure, weak hashing mechanisms, banner disclosures, and unauthenticated control protocols pose significant operational risks..

Key Risk Findings

1. Cleartext Administrative Credential Exposure (High Risk):

HTTP Basic authentication (http.authbasic) transmitted Base64-encoded credentials (admin:[REDACTED] and root:[REDACTED]). Web form submissions (/goform/svLogin) similarly sent unencrypted credentials over POST requests. Any node with access to internal network traffic can capture these credentials to compromise management web consoles.

2. Obsolete Cookie Hashing Mechanisms (High Risk):

HTTP POST interactions with endpoint 192.168.88.61 revealed session cookies storing user passwords as raw MD5 hashes (Password508=3f7caa3d...). MD5 lacks collision resistance and can be cracked via rainbow tables to yield underlying passwords (ADMIN).

3. Asset Infrastructure Information Disclosure (Medium Risk):

FTP service banners broadcast precise hardware and firmware details (220 AXIS 206 Network Camera 4.40). MAC address OUI analysis further identified hardware vendors including Axis Communications, Digi International, Moxa, and Siemens. Attackers can leverage these metadata points to compile target lists for specific CVE exploits.

4. Insecure Industrial Control Protocols (High Risk):

Modbus TCP (Function Codes 1 and 43) and DNP3 communications operate entirely in cleartext without origin authentication or frame integrity validation. An adversary on the local network segment can issue unauthorized control function codes or manipulate field registers.

5. Unauthenticated Siemens S7 PLC Variable Polling (High Risk):

Siemens S7 (s7comm) traffic demonstrated active Read Var operations (0x04) pulling variable states directly from PLCs without session authentication. In an unsegmented network, malicious actors could issue unauthorized write commands to modify PLC logic.